

The complete equidistribution classification of length-three mesh patterns on layered permutations

Jiahao Zhang

College of Mathematical Sciences & Institute of Mathematics and Interdisciplinary
Sciences
Tianjin Normal University
Tianjin 300387, P. R. China

Abstract. Distributional results for short mesh patterns have largely concerned selected pattern families and host classes. On layered permutations, length-two mesh patterns form 35 distribution classes, whereas at length three there are $6 \cdot 2^{16} = 393,216$ mesh patterns. Here we classify all length-three mesh patterns on the same host class. For patterns whose underlying permutation is layered, we introduce a layer-constraint signature and derive occurrence and signature-enumeration formulas valid at arbitrary length. At length three, the signature reduction, pointwise identities, composition bijections, and two nonlocal identities yield exactly 146 nonzero distribution classes. The two non-layered underlying permutations contribute a single zero class, giving 147 classes in total. Each equidistribution used for the upper bound is proved for hosts of every size, and pairwise separation of the 146 representatives gives the matching lower bound.

Keywords: mesh pattern, layered permutation, equidistribution, composition, classification.

2020 Mathematics Subject Classification: 05A05, 05A15.

1 Introduction

Let S_n denote the set of permutations of $[n] = \{1, \dots, n\}$. If $\pi = \pi_1 \cdots \pi_n \in S_n$ and $\tau = \tau_1 \cdots \tau_k \in S_k$, an *occurrence* of the classical pattern τ in π is a sequence of indices $1 \leq i_1 < \cdots < i_k \leq n$ such that

$$\pi_{i_a} < \pi_{i_b} \iff \tau_a < \tau_b \quad (1 \leq a, b \leq k).$$

Thus the selected entries have the same relative order as τ . For example, the subsequence $\pi_1\pi_2\pi_4 = 254$ of $\pi = 25143$ is an occurrence of 132. We refer to Kitaev [7] for general background on classical permutation patterns.

Mesh patterns, introduced by Brändén and Claesson [3], refine this notion by imposing emptiness conditions around a classical occurrence. A mesh pattern of length k is a pair $p = (\tau, R)$, where $\tau \in S_k$ and $R \subseteq [0, k] \times [0, k]$, with $[0, k] = \{0, 1, \dots, k\}$. Fix an occurrence at positions $i_1 < \cdots < i_k$, let $v_1 < \cdots < v_k$ be its selected values in increasing order, and set

$$i_0 = v_0 = 0, \quad i_{k+1} = v_{k+1} = n + 1.$$

It is an occurrence of p when, for every $(a, b) \in R$, the open rectangle $(i_a, i_{a+1}) \times (v_b, v_{b+1})$ contains no point (j, π_j) . In the standard diagram one plots the points (a, τ_a) and shades the unit cells indexed by R . For example,



$$p = (132, \{(0, 0), (1, 1), (1, 2), (3, 1)\}).$$

The entries 243 in the permutation 1243 form a classical occurrence of 132, but not an occurrence of the displayed mesh pattern: the unselected point $(1, 1)$ lies in the shaded cell $(0, 0)$. By contrast, the first three entries of 1324 do form an occurrence, since its only unselected point lies in the unshaded cell $(3, 3)$. To save space, throughout the paper we write the underlying permutation in one-line notation and record the shaded set algebraically; after the layer reduction we use the still more compact signature notation.

Write $p(\pi)$ for the number of occurrences of p in π . For a fixed host family $\mathcal{C}$, define

$$D_{p,n}^{\mathcal{C}}(q) = \sum_{\substack{\pi \in \mathcal{C} \\ |\pi| = n}} q^{p(\pi)}.$$

Two patterns are *distribution equivalent* on $\mathcal{C}$ when these polynomials agree for every n . This is distinct from coincidence, which asks for equality of the avoidance sets, and from Wilf-equivalence, which asks only for equality of their cardinalities. Because a mesh pattern of length k has $(k+1)^2$ cells, there are $k! 2^{(k+1)^2}$ raw patterns. Our problem is to classify the $6 \cdot 2^{16} = 393,216$ patterns of length three by distribution equivalence when $\mathcal{C}$ is the class of layered permutations.

The literature on short mesh patterns began with avoidance. Hilmarsson *et al.* [6] initiated the Wilf-classification at length two, reducing the 1,024 raw patterns to at most 65 candidates and then to an upper bound of 56 Wilf-classes. Kitaev and Zhang [9] began the systematic study of full occurrence distributions for these patterns. Further length-two equidistributions and infinite-family extensions were obtained by Han and Zeng [5] and by Kitaev, Zhang, and Zhang [10]. Related short-pattern statistics have also been pursued on restricted hosts, including alternating permutations [8, 4], while distributions of selected ordinary length-two mesh patterns have been studied on king permutations [11].

At length three, Bean, Gudmundsson, Magnusson, and Ulfarsson [2] completed the coincidence classification. That result determines when two patterns have exactly the same avoiders, but it does not determine their full occurrence distributions. Distributional work at length three has instead concentrated on structured subfamilies, notably joint equidistributions of mesh patterns with underlying permutations 123 and 132 or 123 and 321 and with symmetric or minus-antipodal shadings [12, 13, 14]. Thus, despite substantial progress, these studies do not classify the distributions of all 393,216 length-three mesh patterns.

Layered permutations provide a setting in which a complete result becomes possible. They form a natural and well-studied permutation class [1]; every layered permutation is encoded by a composition, and every classical occurrence has a canonical description in terms of supporting layers. Our recent length-two classification on this class [16] used

that description case by case and obtained 35 distribution classes. To the best of our knowledge, the present paper is the first to show that the full distributional classification of length-three mesh patterns can also be completed on a specified nontrivial host class.

The layer-constraint signature is the main organizing device. The active-cell theorem and the signature formula are valid for mesh patterns of arbitrary length whose underlying permutation is layered. They yield a rational generating function for total occurrences and an explicit enumeration of signatures. At length three, these tools reduce the 393,216 raw diagrams to 1,600 effective shadings and then to 644 signatures. Pointwise identities, reverse-complement, the unique-support classification, layer-coordinate relabellings, and five families of distributional bridges then give 146 nonzero distribution classes. The two non-layered underlying permutations contribute one zero class. Figure 1 records the intermediate quotients, and Theorem 5.1 establishes the matching upper and lower bounds.

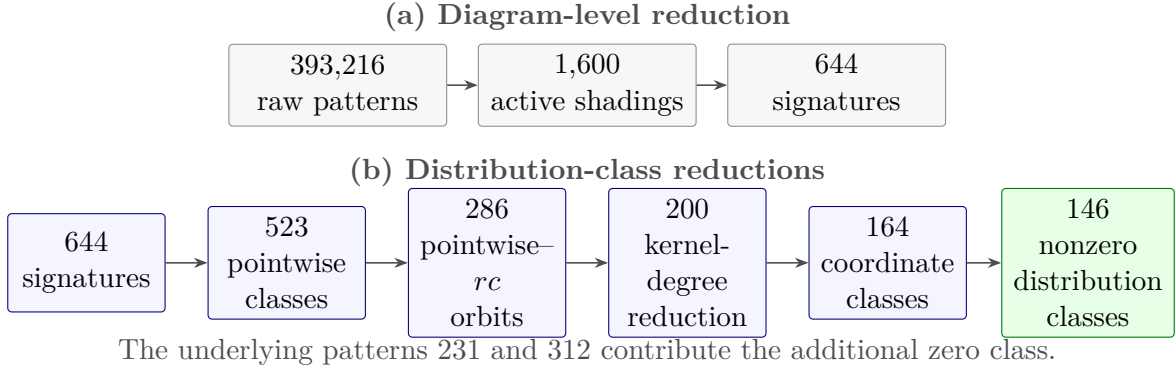


Figure 1: The two levels of the classification. Panel (a) records the diagram-level reduction. Panel (b) gives the distribution-class reductions; its arrows represent, from left to right, pointwise equality, reverse-complement, kernel-degree reduction, layer-coordinate relabelling, and the remaining distributional bridges.

Section 2 identifies the active cells on layered hosts. Section 3 derives the signature formula, its enumerative consequences, and the exact pointwise quotient. Section 4 proves the distributional reductions for hosts of every size, and Section 5 assembles the complete classification. The appendix proves the uniform coordinate identities.

2 Layered hosts and active cells

A permutation is *layered* if it is a direct sum of decreasing permutations. For positive integers $\ell_1, \dots, \ell_m$, write

$$L(\ell_1, \dots, \ell_m) = \delta_{\ell_1} \oplus \dots \oplus \delta_{\ell_m}, \quad \delta_r = r(r-1) \dots 1.$$

The class of all layered permutations is denoted by $\mathcal{L}$. Its members of size n are in bijection with the compositions of n . We henceforth suppress the superscript $\mathcal{L}$ in the distribution polynomial $D_{p,n}^{\mathcal{L}}(q)$.

Every subpermutation of a layered permutation is layered. The layered permutations of length three are

$$123 = \delta_1 \oplus \delta_1 \oplus \delta_1, \quad 132 = \delta_1 \oplus \delta_2, \quad 213 = \delta_2 \oplus \delta_1, \quad 321 = \delta_3.$$

Consequently, the underlying permutations 231 and 312 give the zero statistic, regardless of the shading.

Suppose that the layered underlying permutation has block sizes $r = (r_1, \dots, r_s)$, and put $R_j = r_1 + \dots + r_j$, with $R_0 = 0$. Define the internal channel of block j and the external boundary cells by

$$\mathcal{C}_j = \{(R_{j-1} + h, R_j - h) : 0 \leq h \leq r_j\}, \quad \mathcal{E}_j = (R_j, R_j) \quad (0 \leq j \leq s).$$

Theorem 2.1 (Active cells). *For a mesh pattern whose underlying permutation is $\delta_{r_1} \oplus \dots \oplus \delta_{r_s}$, the cells that can contain an unselected point relative to an occurrence in a layered host are exactly*

$$\mathcal{C}_1 \cup \dots \cup \mathcal{C}_s \cup \{\mathcal{E}_0, \dots, \mathcal{E}_s\}.$$

Shading any other cell has no effect on the occurrence statistic on $\mathcal{L}$.

Proof. Every selected decreasing block lies in one host layer. An unselected point in a supporting layer lies in the corresponding internal channel, because position and value ranks change in opposite directions inside a decreasing layer. An unselected layer before, between, or after the supporting layers lies in the corresponding diagonal boundary cell. This proves that no other cell can be occupied. Conversely, each channel cell is realized by inserting an extra point in the appropriate internal gap, and each boundary cell is realized by inserting an unused host layer in the corresponding external gap. Hence every listed cell is active. $\square$

There are $3 + 2s + 1$ active cells at length three. Figure 2 displays their two types for each layered underlying permutation, and Table 1 records the resulting counts.

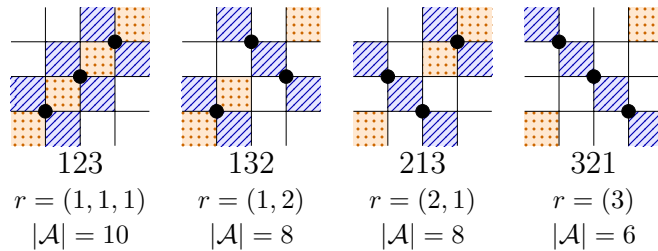


Figure 2: Active cells for the four layered permutations of length three. Blue diagonally hatched cells are internal-channel cells, and orange dotted cells are external boundary cells; unmarked cells are inactive. These markings indicate activity, not mesh shading.

Table 1: Active-cell and signature counts at length three.

τ	block sizes	active cells	effective shadings	signatures
123	(1, 1, 1)	10	$2^{10} = 1,024$	$3^3 2^4 = 432$
132	(1, 2)	8	$2^8 = 256$	$(3 \cdot 4) 2^3 = 96$
213	(2, 1)	8	$2^8 = 256$	$(4 \cdot 3) 2^3 = 96$
321	(3)	6	$2^6 = 64$	$5 \cdot 2^2 = 20$

Thus the four occurring underlying permutations have 1,600 effective shadings. All inactive-cell choices are restored at the end by multiplying with the appropriate power of two.

3 Signatures, enumeration, and pointwise statistics

Let z_j be the number of shaded cells in $\mathcal{C}_j$, and let $E \subseteq \{0, 1, \dots, s\}$ record the shaded external cells. The layer-constraint signature is

$$\Sigma = (r, z, E) = ((r_1, \dots, r_s), (z_1, \dots, z_s), E).$$

For a supporting layer of size ℓ , define

$$K_{r,z}(\ell) = \begin{cases} \binom{\ell - z}{r - z}, & 0 \leq z \leq r, \\ \mathbf{1}_{\ell=r}, & z = r + 1. \end{cases}$$

Here and below, $\binom{a}{b} = 0$ unless $a \geq b \geq 0$; in particular, $K_{r,z}(\ell) = 0$ when $\ell < r$. The first case follows by distributing the $\ell - r$ unselected points among the $r + 1 - z$ unshaded internal gaps. The second case is rigid: all internal gaps must be empty.

For a host composition $L = (\ell_1, \dots, \ell_m)$, let $\Phi_E(m)$ consist of the increasing maps from $\{1, \dots, s\}$ to $\{1, \dots, m\}$ satisfying

$$0 \in E \Rightarrow \phi(1) = 1, \quad s \in E \Rightarrow \phi(s) = m, \quad j \in E \cap \{1, \dots, s-1\} \Rightarrow \phi(j+1) = \phi(j)+1.$$

Thus the elements 0 and s encode only the two endpoint conditions, while the internal elements of E encode adjacency. We set $\Phi_E(m) = \emptyset$ when $m < s$; when $s = 1$ there is no internal adjacency condition.

Theorem 3.1 (Signature formula). *For every layered host $L = (\ell_1, \dots, \ell_m)$,*

$$P_\Sigma(L) = \sum_{\phi \in \Phi_E(m)} \prod_{j=1}^s K_{r_j, z_j}(\ell_{\phi(j)}).$$

Consequently, equal signatures determine equal statistics on every individual layered permutation, so the distributional classification on layered hosts factors through signatures.

Proof. Choose the supporting host layers through ϕ . The shaded external cells give precisely the boundary and adjacency conditions defining $\Phi_E(m)$. Once the supporting layers are fixed, the choices inside distinct layers are independent, and the j th layer contributes K_{r_j, z_j} . Multiplication counts the occurrences with that support, and summation over the admissible supports gives the formula. $\square$

We next record two enumerative consequences of the signature formula. Let

$$\Lambda(t) = 1 + \sum_{n \geq 1} 2^{n-1} t^n = \frac{1-t}{1-2t},$$

the ordinary generating function for compositions, including the empty composition. For a signature $\Sigma = (r, z, E)$, put $f(\Sigma) = s + 1 - |E|$ and

$$\alpha(\Sigma) = \sum_{\substack{1 \leq j \leq s \\ z_j \leq r_j}} (r_j - z_j + 1).$$

Proposition 3.2 (Total-occurrence generating function at arbitrary length). *For every signature Σ of length k ,*

$$T_\Sigma(t) := \sum_{\pi \in \mathcal{L}} P_\Sigma(\pi) t^{|\pi|} = \frac{t^k \Lambda(t)^{f(\Sigma)}}{(1-t)^{\alpha(\Sigma)}}.$$

Proof. Fix an occurrence of Σ . Each unshaded external gap may contain an arbitrary, possibly empty, sequence of unused host layers, and therefore contributes the composition series $\Lambda(t)$. Hence the $f(\Sigma)$ free external gaps contribute $\Lambda(t)^{f(\Sigma)}$.

For a non-rigid supporting coordinate,

$$\sum_{\ell \geq 1} K_{r,z}(\ell) t^\ell = \frac{t^r}{(1-t)^{r-z+1}},$$

whereas a rigid coordinate contributes t^r . Multiplying these contributions over the supporting layers and using $\sum_j r_j = k$ gives the stated formula. $\square$

Proposition 3.3 (Signature enumeration at arbitrary length). *Let S_k be the number of layer-constraint signatures arising from length- k mesh patterns whose underlying permutations are layered. Then*

$$S_k = \sum_{(r_1, \dots, r_s) \models k} 2^{s+1} \prod_{j=1}^s (r_j + 2), \quad \sum_{k \geq 1} S_k x^k = \frac{12x - 8x^2}{1 - 8x + 5x^2}.$$

Equivalently, $S_1 = 12$, $S_2 = 88$, and $S_k = 8S_{k-1} - 5S_{k-2}$ for $k \geq 3$; in particular, $S_3 = 644$ and $S_4 = 4,712$.

Proof. For a fixed block composition $(r_1, \dots, r_s)$, each z_j has $r_j + 2$ possible values and $E \subseteq \{0, \dots, s\}$ has 2^{s+1} possible values. This gives the stated sum. Put

$$A(x) = \sum_{r \geq 1} 2(r+2)x^r = \frac{2x(3-2x)}{(1-x)^2}.$$

Since $2^{s+1} \prod_{j=1}^s (r_j+2) = 2 \prod_{j=1}^s (2(r_j+2))$, summing over all nonempty block compositions gives

$$\sum_{k \geq 1} S_k x^k = 2 \sum_{s \geq 1} A(x)^s = \frac{2A(x)}{1-A(x)} = \frac{12x - 8x^2}{1 - 8x + 5x^2}.$$

Comparing coefficients gives the stated recurrence and initial values. $\square$

The number of effective mesh patterns represented by a signature is $\nu(\Sigma) = \prod_{j=1}^s \binom{r_j+1}{z_j}$. At length three, summing these multiplicities over the $S_3 = 644$ signatures restores the 1,600 effective shadings. Every subsequent classification step is carried out on the 644 signatures.

Equal signatures are not necessary for pointwise equality. Put

$$\mathcal{B} = (1, x, \binom{x}{2}, \binom{x}{3}, \mathbf{1}_{x=1}, \mathbf{1}_{x=2}, \mathbf{1}_{x=3}).$$

Every length-three kernel $K_{r,z}$ has an exact expansion in $\mathcal{B}$. The cutoff functions are important; for example, $K_{2,2}(x) = 1 - \mathbf{1}_{x=1}$ rather than the constant function one.

The seven functions in $\mathcal{B}$ are linearly independent on $\mathbb{Z}_{>0}$. Indeed, a linear relation restricted to $x \geq 4$ forces its polynomial part, of degree at most three, to vanish identically; evaluating the remaining relation at $x = 1, 2, 3$ then forces the three indicator coefficients to vanish. It follows, successively in each coordinate, that $\mathcal{B}^{\otimes m}$ is linearly independent for every m . Hence, for every number m of host layers, the tensor expansion is unique.

Theorem 3.4 (Exact pointwise quotient). *The 644 signatures induce exactly 523 distinct occurrence statistics on individual layered permutations.*

Proof. Let X denote the formal letter corresponding to the constant function 1 in $\mathcal{B}$, and put $G = 1 + X + X^2 + \dots = (1 - X)^{-1}$. Expand each kernel $K_{r,z}$ as the corresponding linear form $\kappa_{r,z}$ in the seven letters of $\mathcal{B}$. The full family of coefficient tensors of a signature is the noncommutative formal series

$$G_0 \kappa_{r_1, z_1} G_1 \cdots \kappa_{r_s, z_s} G_s, \quad G_j = \begin{cases} 1, & j \in E, \\ G, & j \notin E. \end{cases}$$

Indeed, G_j records the unused host layers in the j th external gap.

This expression has a finite canonical form. Expand every $\kappa_{r,z}$ into its X -part and its six remaining letters, and collect terms according to the ordered word of non- X letters. Between two consecutive such letters there remains only a rational series in X , obtained from at most four factors chosen from 1 and G and at most three explicit factors X . For a

fixed word of non- X letters, introduce one variable for each intervening run and put every denominator over $(1 - X_i)^4$; the resulting polynomial numerator is unique. Equality is therefore decided by comparing a finite list of integer numerator polynomials. Reducing the 644 signatures coefficientwise in this manner yields 523 distinct canonical forms. Equal forms give identical tensors for every m , whereas distinct forms differ in at least one tensor coefficient. The independence of $\mathcal{B}^{\otimes m}$ then proves the stated quotient over all layered hosts. $\square$

4 From signatures to distributions

4.1 Reverse-complement after the pointwise quotient

Reverse-complement rotates a permutation plot by 180 degrees and preserves the class $\mathcal{L}$. At the signature level it acts by

$$(r_1, \dots, r_s; z_1, \dots, z_s; E) \mapsto (r_s, \dots, r_1; z_s, \dots, z_1; s - E),$$

where $s - E = \{s - e : e \in E\}$. Inversion fixes the signature: it only reverses the cells inside each internal channel and therefore preserves their number z_j , while every external cell lies on the diagonal. Thus inversion gives no additional quotient after signatures have been formed.

Pointwise equality is preserved by reverse-complement, so this involution descends from signatures to the 523 pointwise classes of Theorem 3.4.

Proposition 4.1 (Pointwise reverse-complement quotient). *The 523 pointwise classes form 286 pointwise-rc orbits.*

Proof. Among the canonical formal-series representatives of Theorem 3.4, exactly 49 are fixed by the induced involution. The remaining 474 form 237 pairs, so Burnside's lemma gives $49 + 474/2 = 286$. $\square$

4.2 Kernel-degree reduction for unique supports

Let $f(\Sigma) = s + 1 - |E|$ be the number of free external gaps. The range $f \leq 1$ is special. At length three the possible values of f are

underlying pattern	s	possible values of f
123	3	0, 1, 2, 3, 4
132, 213	2	0, 1, 2, 3
321	1	0, 1, 2.

When $f \leq 1$, all unused host layers must lie in the unique free external gap, if such a gap exists. Consequently every host admitting the prescribed core has a unique support injection. If $f \geq 2$, by contrast, unused layers can be distributed among several free gaps,

and the occurrence statistic becomes a sum over several possible supports. The resulting global support geometry is not captured by the unordered local data below; these multiple-support cases are treated in the next two subsections.

For the unique-support range, write a supporting layer as $\ell_j = r_j + u_j$. For every non-rigid coordinate put $d_j = r_j - z_j$ and define the degree multiset $D(\Sigma) = \{\{d_j : z_j \leq r_j\}\}$. Rigid coordinates are omitted because they force $u_j = 0$ and contribute the factor one. The local occurrence number becomes

$$\omega_D(\mathbf{u}) = \prod_{d_j \in D} \binom{u_j + d_j}{d_j}.$$

For $f(\Sigma) \leq 1$, define

$$F_\Sigma(t, q) = \sum_{\pi \in \mathcal{L}} q^{P_\Sigma(\pi)} t^{|\pi|},$$

$$C_D(t, q) = t^3 \sum_{\mathbf{u} \geq 0} t^{|\mathbf{u}|} (q^{\omega_D(\mathbf{u})} - 1).$$

Proposition 4.2 (Distribution series in the unique-support range). *If $f(\Sigma) \leq 1$, then*

$$F_\Sigma(t, q) = \begin{cases} \Lambda(t) + C_D(t, q), & f(\Sigma) = 0, \\ \Lambda(t)(1 + C_D(t, q)), & f(\Sigma) = 1. \end{cases}$$

Proof. Start with the contribution $\Lambda(t)$ of all layered hosts, each carrying weight one. A host with a nonzero occurrence contains the prescribed core with a unique support. Replacing its baseline weight 1 by $q^{\omega_D(\mathbf{u})}$ gives the correction $C_D(t, q)$ when there is no free external gap. When there is one free external gap, that gap contains an arbitrary composition and contributes the additional factor $\Lambda(t)$. $\square$

Corollary 4.3 (Avoidance in the unique-support range). *Let*

$$A_\Sigma(t) = \sum_{\substack{\pi \in \mathcal{L} \\ P_\Sigma(\pi) = 0}} t^{|\pi|}.$$

If $f(\Sigma) \leq 1$, then

$$A_\Sigma(t) = \Lambda(t) - \frac{t^3 \Lambda(t)^{f(\Sigma)}}{(1 - t)^{|D(\Sigma)|}}.$$

Consequently, in the unique-support range the avoidance generating function depends only on $f(\Sigma)$ and $|D(\Sigma)|$.

Theorem 4.4 (Unique-support classification). *Let Σ and Σ' be length-three signatures with at most one free external gap. Then they are equidistributed on $\mathcal{L}$ if and only if*

$$f(\Sigma) = f(\Sigma') \quad \text{and} \quad D(\Sigma) = D(\Sigma').$$

Proof. For sufficiency, Proposition 4.2 shows directly that the distribution series depends only on (f, D) .

For necessity, suppose that Σ and Σ' are equidistributed. By Proposition 3.2, their total-occurrence series are equal. Since $f \in \{0, 1\}$, the pole at $t = 1/2$ determines f , and hence $f(\Sigma) = f(\Sigma')$.

After removing the common free-gap factor, Proposition 4.2 implies that the corresponding correction series $C_D(t, q)$ agree. The universal initial term is $t^3(q - 1)$. The first subsequent coefficient that depends on D is

$$[t^4]C_D(t, q) = \sum_{d \in D} (q^{d+1} - 1),$$

because exactly one surplus variable equals one. This polynomial records every element of D , including its multiplicity. Hence $D(\Sigma) = D(\Sigma')$. $\square$

Remark 4.5 (Why the unique-support hypothesis is sharp). *Consider*

$$\Sigma_1 = (111; 000; \{0, 1\}), \quad \Sigma_2 = (111; 000; \{0, 2\}).$$

Both signatures have $f = 2$ and $D = \{\{1, 1, 1\}\}$, but their support geometries are different. On the host composition $L = (1, 1, 2, 2)$, the first signature permits the third support to be either layer 3 or layer 4, whereas the second has admissible supports $(1, 2, 3)$ and $(1, 3, 4)$. Therefore

$$\begin{aligned} P_{\Sigma_1}(L) &= \ell_1 \ell_2 (\ell_3 + \ell_4) = 4, \\ P_{\Sigma_2}(L) &= \ell_1 \ell_2 \ell_3 + \ell_1 \ell_3 \ell_4 = 6. \end{aligned}$$

Thus

$$D_{\Sigma_1, 6}(q) = 6 + 10q^4 + 14q^6 + 2q^8, \quad D_{\Sigma_2, 6}(q) = 6 + 9q^4 + 2q^5 + 13q^6 + 2q^8.$$

Hence the two signatures are not equidistributed. This is a concrete instance of the lower-bound separation used in Theorem 5.1.

At length three the possible degree multisets are

$$\begin{aligned} \emptyset, \quad \{0\}, \{1\}, \{2\}, \{3\}, \quad \{0, 0\}, \{0, 1\}, \{0, 2\}, \{1, 1\}, \{1, 2\}, \\ \{0, 0, 0\}, \{0, 0, 1\}, \{0, 1, 1\}, \{1, 1, 1\}. \end{aligned}$$

The 56 fully constrained signatures therefore form exactly fourteen distribution classes, and the 190 signatures with one free gap form another fourteen distribution classes. Hence 246 signatures are classified into 28 distribution classes. In the pointwise- rc quotient this replaces 114 pointwise- rc orbits by 28 distribution classes and reduces the current upper bound from 286 to

$$286 - 114 + 28 = 200.$$

For example, the three local data

$$(r, z) = (3, 2), \quad (r, z) = ((1, 2), (0, 3)), \quad (r, z) = ((1, 1, 1), (0, 2, 2))$$

all have degree multiset $D = \{1\}$, despite arising from different underlying layered permutations.

4.3 Layer-coordinate relabelling

Theorem 4.4 completes the unique-support range $f \leq 1$. We now consider the pointwise- rc orbits represented by the remaining signatures with $f \geq 2$. In these multiple-support cases a host may admit several support injections, so $P_\Sigma(L)$ is a sum of kernel products, and the unordered degree multiset $D(\Sigma)$ no longer records how the summands are positioned among the host layers.

Some of these sums nevertheless agree after a permutation of the variables $\ell_1, \dots, \ell_m$. Such a relabelling acts on the host composition $L = (\ell_1, \dots, \ell_m)$, not on the mesh pattern. For fixed n and m it is a size-preserving bijection on the compositions of n with m parts, and therefore gives the following composition-level criterion.

Theorem 4.6 (Layer-coordinate criterion). *Suppose that for every m there is $\rho_m \in S_m$ such that*

$$P_\Sigma(\ell_1, \dots, \ell_m) = P_{\Sigma'}(\ell_{\rho_m(1)}, \dots, \ell_{\rho_m(m)})$$

for all positive ℓ_i . Then Σ and Σ' are equidistributed on $\mathcal{L}$.

Proof. The coordinate permutation is a bijection of the compositions of each fixed n and m . The displayed identity therefore gives the same multiset of occurrence values. Summing over m proves equality of the distribution polynomials for every n . $\square$

Unlike Theorem 4.4, the layer-coordinate criterion is only sufficient: it does not assert that every multiple-support equidistribution arises from a coordinate permutation. It supplies uniform bijections for the identifications recorded below, while the cases not reached by such permutations require the composition bijections of Section 4.4.

For example, put $\Sigma_A = (111; 000; \{0, 1\})$ and $\Sigma_B = (111; 000; \{0, 3\})$. For Σ_A , the constraints $0, 1 \in E$ force the first two supporting layers to be layers 1 and 2, while the third support may be any later layer. For Σ_B , the first and third supports are layers 1 and m , while the middle support may be any intermediate layer. Since every local kernel is $K_{1,0}(\ell) = \ell$, the two occurrence polynomials are

$$P_{\Sigma_A}(L) = \ell_1 \ell_2 \sum_{j=3}^m \ell_j \quad \text{and} \quad P_{\Sigma_B}(L) = \ell_1 \ell_m \sum_{j=2}^{m-1} \ell_j.$$

Under the coordinate bijection

$$L = (\ell_1, \ell_2, \ell_3, \dots, \ell_m) \mapsto L' = (\ell_1, \ell_3, \dots, \ell_m, \ell_2),$$

we have $P_{\Sigma_B}(L') = \ell_1 \ell_2 \sum_{j=3}^m \ell_j = P_{\Sigma_A}(L)$. Thus the two statistics are equidistributed even though their support geometries differ and the signatures are not related by reverse-complement.

The uniform coordinate permutations are listed in Appendix A. There Lemma A.1 proves their identities for arbitrary m .

After the kernel-degree reduction, 172 multiple-support pointwise- rc orbits remain. Eighty-four are unchanged by the coordinate identities, while the 43 identities generate

52 equivalence classes on the other 88 orbits. Supplementary Table S2 gives the source signature, target signature, orientation, and coordinate word for every identity. The resulting upper bound is therefore

$$28 + 84 + 52 = 164.$$

4.4 The remaining distributional bridges

At this point the local kernel data and the coordinate symmetries have been exhausted. The remaining identifications act on host compositions themselves: they move cuts, contract marked adjacent blocks, complement separator words, or invoke a nonlocal distributional identity.

Write a layered host as $L = (\ell_1, \dots, \ell_m)$. The signature formula expresses the occurrence count as a sum over supports of products of local kernel weights. The remaining identities can therefore be formulated as size-preserving maps on compositions and their marked supports.

We encode L by its separator word

$$w(L) = 0^{\ell_1-1} 1 0^{\ell_2-1} 1 \dots 1 0^{\ell_m-1};$$

the ones mark the cuts between consecutive parts.

For a signature Σ and a host composition L , let $\text{Occ}_\Sigma(L)$ denote the set of occurrences of Σ in L . For $k \geq 0$, a k -marked object is a pair (L, M) with $M \subseteq \text{Occ}_\Sigma(L)$ and $|M| = k$.

Lemma 4.7 (Marked-occurrence principle). *Suppose that, for every n and $k \geq 0$, there is a size-preserving bijection between the k -marked objects of size n for Σ and those for Σ' . Then Σ and Σ' are equidistributed on $\mathcal{L}$.*

Proof. For fixed n and k , put

$$B_{\Sigma,n,k} = \sum_{L \models n} \binom{P_\Sigma(L)}{k}.$$

This is exactly the number of k -marked objects of size n . Hence the assumed bijections give $B_{\Sigma,n,k} = B_{\Sigma',n,k}$ for every $k \geq 0$. Using $q^r = (1 + (q - 1))^r = \sum_{k \geq 0} \binom{r}{k} (q - 1)^k$, we obtain

$$\begin{aligned} D_{\Sigma,n}(q) &= \sum_{L \models n} q^{P_\Sigma(L)} \\ &= \sum_{k \geq 0} B_{\Sigma,n,k} (q - 1)^k. \end{aligned}$$

The same expansion holds for Σ' . Since the binomial moments $B_{\Sigma,n,k}$ and $B_{\Sigma',n,k}$ agree for every k , the two distribution polynomials are equal for every n . $\square$

The bridges fall into five families. Three are direct bijections on compositions—cut shifts, contractions, and separator complementation. The other two use a rational generating function and MacMahon’s equidistribution, respectively.

Proposition 4.8 (Uniform cut-shift bijections). *For $a, b \in \{0, 1, 2\}$, let*

$$\Sigma_{a,b} = (111; (a, b, 2); \{0, 3\}), \quad \Sigma'_{a,b} = (12; (b, a + 1); \{2\}).$$

Then $\Sigma_{a,b}$ and $\Sigma'_{a,b}$ are equidistributed on $\mathcal{L}$.

Proof. For a host composition $L = (\ell_1, \dots, \ell_m)$, the signature formula gives

$$P_{\Sigma_{a,b}}(L) = \mathbf{1}_{\ell_m=1} K_{1,a}(\ell_1) \sum_{j=2}^{m-1} K_{1,b}(\ell_j).$$

Thus every host with positive occurrence number has the form $L = (x, \beta, 1)$, where β is a nonempty composition. Define $\Phi(x, \beta, 1) = (\beta, x + 1)$. For the target signature,

$$P_{\Sigma'_{a,b}}(\beta, x + 1) = K_{2,a+1}(x + 1) \sum_{c \in \beta} K_{1,b}(c).$$

The kernel identity $K_{2,a+1}(x + 1) = K_{1,a}(x)$ for $a = 0, 1, 2$ therefore gives $P_{\Sigma_{a,b}}(x, \beta, 1) = P_{\Sigma'_{a,b}}(\Phi(x, \beta, 1))$.

The map is size preserving, and its inverse on positive-occurrence hosts is $(\beta, y) \mapsto (y - 1, \beta, 1)$ for $y \geq 2$. Hence, for every n and every positive occurrence number, the corresponding hosts are in bijection. Since the total number of compositions of n is the same on both sides, the zero-occurrence hosts also occur equally often. Thus the two signatures are equidistributed. $\square$

Proposition 4.9 (Exceptional rational bridge). *The signatures*

$$\Sigma = (111; (1, 2, 1); \{0, 3\}), \quad \Sigma' = (12; (2, 1); \{1\})$$

are equidistributed on $\mathcal{L}$.

Proof. For a composition $L = (\ell_1, \dots, \ell_m)$, the first statistic is $P_{\Sigma}(L) = \#\{j : 2 \leq j \leq m - 1, \ell_j = 1\}$, the number of interior singleton parts. Put $A(t) = t/(1 - t)$ and $I(t, q) = qt + t^2/(1 - t)$. The first and last parts are arbitrary, while every interior part contributes a factor q precisely when it is a singleton. Hence

$$\begin{aligned} G_{\Sigma}(t, q) &:= \sum_L q^{P_{\Sigma}(L)} t^{|L|} \\ &= 1 + A(t) + \frac{A(t)^2}{1 - I(t, q)} \\ &= \frac{1 - qt}{1 - (1 + q)t + (q - 1)t^2}. \end{aligned}$$

For the second signature,

$$P_{\Sigma'}(L) = \sum_{i=1}^{m-1} \mathbf{1}_{\ell_i=1} (\ell_{i+1} - 1).$$

Let H_0 be the generating function for a suffix when the preceding part is not a singleton, or when there is no preceding part, and let H_1 be the corresponding series when the preceding part is a singleton. Decomposing according to the next part gives

$$\begin{aligned} H_0 &= 1 + tH_1 + \frac{t^2}{1-t}H_0, \\ H_1 &= 1 + tH_1 + \frac{t^2q}{1-tq}H_0. \end{aligned}$$

Solving these equations yields

$$H_0 = \frac{1-qt}{1-(1+q)t+(q-1)t^2} = G_\Sigma(t, q).$$

Since H_0 is the distribution series for Σ' , the two signatures are equidistributed. $\square$

Proposition 4.10 (Unified contraction bridges). *For $z \in \{0, 1, 2, 3\}$, put*

$$\Sigma_z = (12; (2, z); \{1\}), \quad \Sigma'_z = (3; z+1; \emptyset).$$

For each z , Σ_z and Σ'_z are equidistributed on $\mathcal{L}$.

Proof. Because the first local parameter is $2 = r_1 + 1$, the first supporting layer is rigid. Hence every occurrence of Σ_z is supported on an adjacent pair $(1, b)$ with $b \geq 2$. Two distinct such support pairs cannot overlap: the second member of a support pair has size at least two and therefore cannot be the singleton first member of another support pair.

Consider a k -marked object. Group the marks according to their supporting pairs and simultaneously contract every support pair carrying at least one mark by $(1, b) \mapsto b+1$. The identity $K_{2,z}(b) = K_{3,z+1}(b+1)$ shows that the local occurrence sets on the two sides have the same cardinality. For $z \leq 2$, both may be canonically indexed by the subsets $\binom{[b-z]}{2-z}$, and for $z = 3$ both local occurrence sets are singletons. We transfer the marked local occurrences under this identification.

The inverse is equally explicit. Every target part carrying at least one mark has size $b+1 \geq 3$; replace it by $(1, b)$ and transfer its marked local occurrences back under the same identification. Unmarked parts and unmarked support pairs are left unchanged.

Thus we obtain a size-preserving bijection between the k -marked objects for Σ_z and Σ'_z for every k . The result follows from Lemma 4.7. $\square$

Proposition 4.11 (Uniform separator-complement bridges). *For $a \in \{0, 1, 2\}$, let*

$$\Sigma_a = (111; (a, 1, 1); \{0, 1\}), \quad \Sigma'_a = (12; (a, 1); \{0\}).$$

For $L = (\ell_1, \dots, \ell_m)$ with $m \geq 2$, one has

$$P_{\Sigma_a}(L) = K_{1,a}(\ell_1)(m-2), \quad P_{\Sigma'_a}(L) = K_{1,a}(\ell_1) \sum_{j=2}^m (\ell_j - 1).$$

If $m = 1$, both statistics are zero.

Proof. For $m \geq 2$, the signature formula gives the two displayed formulas in the statement; for $m = 1$, both statistics are zero. When $m \geq 2$, write $w(L) = 0^{\ell_1-1}1v$ and define $\Psi(w) = 0^{\ell_1-1}1\bar{v}$, where $\bar{v}$ is obtained from v by exchanging 0 and 1. For compositions with one part, let Ψ be the identity.

The map Ψ is a size-preserving involution and leaves ℓ_1 unchanged. The numbers of 1's and 0's in v are $m - 2$ and $\sum_{j=2}^m (\ell_j - 1)$, respectively. Complementation exchanges these two quantities, so $P_{\Sigma_a}(L) = P_{\Sigma'_a}(\Psi(L))$. Thus Ψ is a size-preserving bijection proving the desired equidistribution. $\square$

Proposition 4.12 (MacMahon bridge). *The signatures $\Sigma = (12; 02; \emptyset)$ and $\Sigma' = (12; 11; \emptyset)$ are equidistributed on $\mathcal{L}$.*

Proof. For the separator word w of a composition $L = (\ell_1, \dots, \ell_m)$, we prove

$$P_{\Sigma}(L) = \sum_{\substack{j \geq 2 \\ \ell_j \geq 2}} (\ell_1 + \dots + \ell_{j-1}) = \text{maj}(w),$$

$$P_{\Sigma'}(L) = \sum_{j=2}^m (j-1)(\ell_j - 1) = \text{inv}(w).$$

Write the separator word as

$$w = 0^{\ell_1-1}10^{\ell_2-1}1 \dots 10^{\ell_m-1}.$$

For every $j \geq 2$ with $\ell_j \geq 2$, the separator immediately preceding the j th part occurs at position $\ell_1 + \dots + \ell_{j-1}$ and is followed by a zero. Hence w has a descent 10 precisely at these positions, and every descent arises in this way. This proves the first displayed identity.

On the other hand, each of the $\ell_j - 1$ zeros belonging to the j th part has exactly $j - 1$ separator ones preceding it, which proves the second displayed identity. MacMahon's equidistribution of major index and inversion number on binary words of fixed content [15] therefore proves the result. $\square$

The five families contain $9 + 1 + 4 + 3 + 1 = 18$ bridges. Comparing their endpoints with the 164 coordinate classes shows that each identifies two previously distinct classes. They therefore reduce the number of classes from 164 to 146.

5 The complete classification

Theorem 5.1 (Complete length-three classification). *The length-three mesh patterns whose underlying permutations are layered form exactly 146 nonzero distribution classes on $\mathcal{L}$. All patterns with underlying permutation 231 or 312 form one additional zero distribution class. Consequently, all $6 \cdot 2^{16}$ length-three mesh patterns form exactly 147 distribution classes on $\mathcal{L}$.*

Proof. For the upper bound, Theorems 2.1 and 3.1 reduce the four occurring underlying permutations first to 1,600 effective shadings and then to 644 signatures. The finite incidence and separation details used below are collected in Supplementary Tables S1–S2 and Proposition S1. The exact pointwise quotient of Theorem 3.4 gives 523 classes, and Proposition 4.1 gives 286 pointwise- rc orbits. The unique-support classification replaces 114 of these orbits by 28 classes, giving 200. The coordinate identities reduce this number to 164. Finally, the eighteen bridges of Section 4.4 reduce it to 146. All identifications in this upper-bound argument are valid for hosts of every size.

For the lower bound, choose one representative signature from each of the 146 candidate classes. Proposition S1 evaluates the occurrence formula of Theorem 3.1 and proves that every two distinct representatives have different distribution polynomials at some size $n \leq 7$. A difference at one size precludes equidistribution, so the 146 representatives lie in distinct classes. There are therefore at least 146 nonzero distribution classes. Every mesh pattern whose underlying permutation is layered occurs in its underlying permutation itself, since there are no unselected points, so none of these classes is the zero class. Finally, 231 and 312 cannot occur in a layered host, and all of their shadings give the one zero distribution class. $\square$

Finally, weighting each signature by its multiplicity $\nu(\Sigma)$ and reinstating the inactive-cell choices accounts for all raw mesh patterns.

6 Discussion

The classification separates three different quotients. Active-cell reduction identifies mesh diagrams that cannot be distinguished on layered hosts. The signature formula identifies statistics that agree on every individual host. Equidistribution is a further quotient and may arise from a nontrivial bijection of compositions even when the pointwise statistics are different.

The signature is more than a change of notation: it converts the geometry of a mesh diagram into a short weighted support polynomial. This polynomial reveals kernel degrees, coordinate symmetries, cut shifts, and separator-word statistics, organizing the 146 classes through a small set of structural reductions.

Within the length-three classification, the MacMahon bridge is the first genuinely non-local word identity: it invokes the classical equidistribution of major index and inversion number on binary words. The same framework extends to larger lengths; Proposition 3.3 already gives $S_4 = 4,712$, although additional word statistics and overlapping marked blocks may arise.

Acknowledgments

The author was supported by the Tianjin Municipal Natural Science Foundation (No. 25JCYBJC00430).

References

- [1] M. Albert, M. Engen, J. Pantone, and V. Vatter, Universal layered permutations, *Electron. J. Combin.* **25** (2018), Paper No. 3.23.
- [2] C. Bean, B. Gudmundsson, T. K. Magnusson, and H. Úlfarsson, Algorithmic coincidence classification of mesh patterns, *Inform. and Comput.* **292** (2023), Article 105014.
- [3] P. Brändén and A. Claesson, Mesh patterns and the expansion of permutation statistics as sums of permutation patterns, *Electron. J. Combin.* **18** (2011), Paper No. P5.
- [4] T. Han, S. Kitaev, and P. B. Zhang, Distribution of maxima and minima statistics on alternating permutations, Springer numbers, and avoidance of flat POPs, *J. Combin. Theory Ser. A* **213** (2025), Article 106034.
- [5] B. Han and J. Zeng, Equidistributions of mesh patterns of length two and Kitaev and Zhang’s conjectures, *Adv. Appl. Math.* **127** (2021), Article 102149.
- [6] I. Hilmarsson, I. Jónsdóttir, S. Sigurdardóttir, L. Vidarsdóttir, and H. Úlfarsson, Wilf-classification of mesh patterns of short length, *Electron. J. Combin.* **22** (2015), Paper No. P4.13.
- [7] S. Kitaev, *Patterns in Permutations and Words*, Springer, Berlin, 2011.
- [8] S. Kitaev and J. Remmel, Quadrant marked mesh patterns in alternating permutations II, *J. Combin.* **4** (2013), 31–65.
- [9] S. Kitaev and P. B. Zhang, Distributions of mesh patterns of short lengths, *Adv. Appl. Math.* **110** (2019), 1–32.
- [10] S. Kitaev, P. B. Zhang, and X. Zhang, Distributions of several infinite families of mesh patterns, *Appl. Math. Comput.* **372** (2020), Article 124984.
- [11] D. Li and P. B. Zhang, Distributions of mesh patterns of short lengths on king permutations, *Discrete Math.* **349** (2026), Article 114681.
- [12] S. Lv and S. Kitaev, On (joint) equidistributions of mesh patterns 123 and 132 with symmetric shadings, *Adv. Appl. Math.* **166** (2025), Article 102856.
- [13] S. Lv and P. B. Zhang, Joint equidistributions of mesh patterns 123 and 132 with minus antipodal shadings, *Discrete Appl. Math.* **379** (2026), 419–433.
- [14] S. Lv and P. B. Zhang, Joint equidistributions of mesh patterns 123 and 321 with symmetric and minus-antipodal shadings, *Ann. Comb.* (2026), [doi:10.1007/s00026-025-00793-8](https://doi.org/10.1007/s00026-025-00793-8).
- [15] P. A. MacMahon, *Combinatory Analysis*, Vol. 2, Cambridge University Press, Cambridge, 1916.

- [16] J. Zhang, S. Kitaev, A. Pyatkin, and X. Su, Equidistribution of mesh patterns of short length on layered permutations, preprint, 2026.

A Uniform layer-coordinate relabellings

All coordinate relabellings needed here are drawn from the following eleven uniform coordinate permutations. A tuple $U = (u_1, \dots, u_m)$ denotes the map $U(\ell_1, \dots, \ell_m) = (\ell_{u_1}, \dots, \ell_{u_m})$. The notation $a:b$ denotes the consecutive string from a to b , in increasing or decreasing order as appropriate; in particular, $2:m = 2, 3, \dots, m$. Thus $U_1 = (1, 3:m, 2)$ means

$$(\ell_1, \ell_2, \ell_3, \dots, \ell_m) \mapsto (\ell_1, \ell_3, \dots, \ell_m, \ell_2).$$

Every local orbit occurring in the 43 connections consists of signatures with three supporting coordinates. Their statistics vanish when $m < 3$, and for these two sizes we take every U_i to be the identity. The displayed coordinate words apply for $m \geq 3$. Each U_i is uniform in m , and Lemma A.1 verifies the identities required by Theorem 4.6.

$$\begin{array}{ll} U_1 & \left| \begin{array}{l} (1, 3:m, 2) \\ (1, m, 2:m-1) \end{array} \right. \\ U_3 & \left| \begin{array}{l} (2, 1, 3:m) \\ (2:m, 1) \end{array} \right. \\ U_5 & \left| \begin{array}{l} (m-1:1, m) \\ (1:m-2, m, m-1) \end{array} \right. \\ U_7 & \left| \begin{array}{l} (2:m-1, 1, m) \\ (m-1, m, 1:m-2) \end{array} \right. \\ U_9 & \left| \begin{array}{l} (m-1, 1:m-2, m) \\ (1, m, m-1:2) \end{array} \right. \\ U_{11} & \left| \begin{array}{l} (3:m, 1, 2) \\ \cdot \end{array} \right. \end{array}$$

Lemma A.1 (Uniform coordinate identities). *Each of the 43 source–target connections listed explicitly in Supplementary Table S2 satisfies, for every m and every positive host composition L ,*

$$P_\Sigma(L) = P_{\Sigma'}(U_i(L)),$$

with the orientation specified by that connection. Thus all these connections satisfy the hypothesis of Theorem 4.6.

Proof. All signatures involved have $r = (1, 1, 1)$. Put

$$b_0(x) = x, \quad b_1(x) = 1, \quad b_2(x) = \mathbf{1}_{x=1}.$$

For $\Sigma = (111; z_1 z_2 z_3; E)$, form the multiset

$$\mathcal{W}_m(\Sigma) = \biguplus_{\phi \in \Phi_E(m)} \{ \{(\phi(j), z_j) : z_j \neq 1\} \}.$$

It records the labeled supporting coordinates after the factors $b_1 = 1$ have been deleted, retaining multiplicity when several supports produce the same reduced word. The signature formula becomes

$$P_\Sigma(L) = \sum_{W \in \mathcal{W}_m(\Sigma)} \prod_{(a,d) \in W} b_d(\ell_a).$$

If $U = (u_1, \dots, u_m)$, let U_*W replace every label (a, d) by (u_a, d) . Consequently,

$$\mathcal{W}_m(\Sigma) = U_*\mathcal{W}_m(\Sigma') \quad (\text{A.1})$$

implies the required identity for every choice of the positive variables $\ell_1, \dots, \ell_m$.

For a listed connection, apply its coordinate word U_i to the admissible supports. If no b_1 factor is deleted, this gives a bijection between the two sets of support triples. If one or two b_1 factors are deleted, the multiplicity of a reduced word is the number of ways to insert the deleted coordinates in the gaps prescribed by E . The conditions defining $\Phi_E(m)$ specify only open or closed gaps, fixed endpoints, and adjacency of consecutive supports. It therefore suffices to verify that U_i maps the corresponding insertion intervals bijectively, preserving their endpoints and multiplicities. The numbers of connections using $U_1, \dots, U_{11}$ are, respectively, 6, 5, 5, 3, 2, 6, 4, 4, 4, 2, 2, which sum to 43.

The example in Section 4.3 covers the case with no deleted factor. The following two examples illustrate one and two deleted factors. For $U_1 = (1, 3:m, 2)$,

$$P_{(111;012;\{0,1\})}(L) = \ell_1 \sum_{k=3}^m \mathbf{1}_{\ell_k=1} = P_{(111;021;\{0,2\})}(U_1(L)),$$

where one constant factor has been deleted. With two constant factors,

$$P_{(111;112;\{0\})}(L) = \sum_{k=3}^m (k-2) \mathbf{1}_{\ell_k=1} = P_{(111;121;\{2\})}(U_1(L)).$$

Every other connection differs from one of these calculations by a cyclic shift, an endpoint interchange, or a reversal of the relevant interval. These operations are bijections of the full symbolic interval, so (A.1) holds for every m . When $m < 3$, both statistics vanish, as noted above. $\square$

These 43 identities reduce the 88 relevant local orbits to 52 classes. Together with the 28 unique-support classes and 84 unaffected singleton multiple-support orbits, this gives $28 + 84 + 52 = 164$.